

Credibility, Strategic Communication, and Information Design

EC 2024 Tutorial

Elliot Lipnowski, Columbia

Doron Ravid, UChicago

PREVIOUS LECTURE AND THIS ONE

Part 1:

(a) Full-commitment case

- “Bayesian Persuasion” (Kamenica Gentzkow 2011)

(b) No-commitment case: state-independent S preferences

- “Cheap Talk with Transparent Motives” (Lipnowski Ravid 2020)

Part 2:

- Models of communication with partial credibility
- Focus on models amenable to belief-based approach

THE PLAN FOR PART 2

(a) A parametric model of **partial commitment**

- “Persuasion via Weak Institutions” (Lipnowski Ravid Shishkin 2022)
- **Stochastic commitment**
- **State-independent S preferences** $\rightsquigarrow$ belief-based approach

(b) Other **qualitative** notions of partial commitment.

- **Manipulable correlation**
- **Mediated communication**
- **Long cheap talk**
- ... more

Stochastic commitment

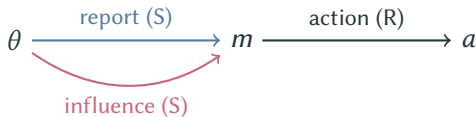
A PARAMETRIC MODEL OF PARTIAL-COMMITMENT COMMUNICATION

Model from three papers

- Bayesian Persuasion under Partial Commitment (Min 2021)
- Rules and Commitment in Communication: An Experimental Analysis (Fréchette Lizzeri Perego 2022)
- Persuasion via Weak Institutions (Lipnowski Ravid Shishkin 2022)

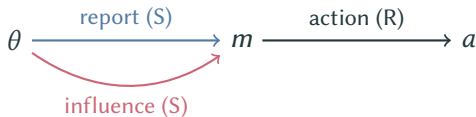
Idea: S can commit **with some probability**

PERSUASION VIA WEAK INSTITUTIONS, JPE 2022



- S publicly announces official report $\xi : \Theta \rightarrow \Delta M$
- S privately observes state $\theta \in \Theta$
- Report m is
 - drawn from $\xi(\cdot|\theta)$ with probability χ
 - freely chosen by Sender with probability $1 - \chi$
- R sees m (but not its source) and chooses $a \in A$
- Again, S has state-independent preferences

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$\chi = 1$ is Bayesian persuasion, $\chi = 0$ is cheap talk

SENDER-OPTIMAL PAYOFF

Geometric characterization

Quasiconcave envelope for messages sent by influencing S

Use concavification for messages *only* in official report

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Non-separable:

- Official messages can't be too tempting to S
- Cap, then concavify

SENDER-OPTIMAL PAYOFF: GEOMETRIC CHARACTERIZATION

Let:

- $\bar{v}(\cdot)$ denote the quasiconcave envelope of $v(\cdot)$
- $v_{\wedge Y}(\cdot)$ denote $\min \{v(\cdot), \bar{v}(Y)\}$
- $\hat{v}_{\wedge Y}(\cdot)$ denote the concave envelope of $v_{\wedge Y}(\cdot)$

SENDER-OPTIMAL PAYOFF: GEOMETRIC CHARACTERIZATION

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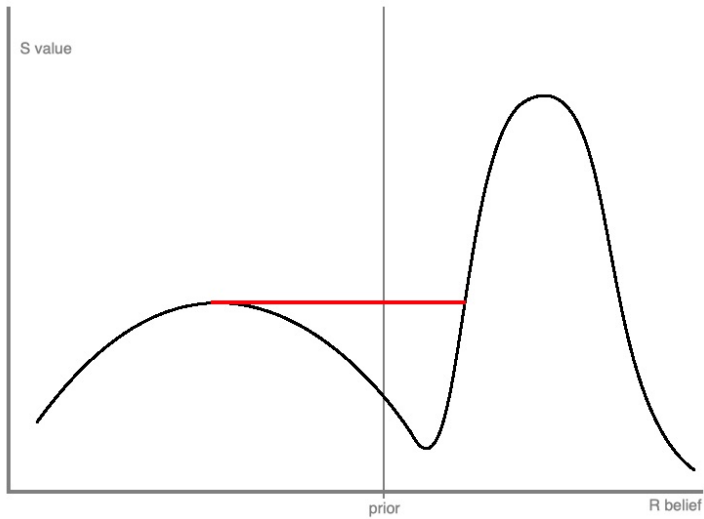
- $\bar{v}(\cdot)$ denote the quasiconcave envelope of $v(\cdot)$
- $v_{\wedge \gamma}(\cdot)$ denote $\min \{v(\cdot), \bar{v}(\gamma)\}$
- $\hat{v}_{\wedge \gamma}(\cdot)$ denote the concave envelope of $v_{\wedge \gamma}(\cdot)$

$$v_{\chi}^*(\mu_0) := \max_{\beta, \gamma \in \Delta\Theta, k \in [0, 1]} \left\{ k \hat{v}_{\wedge \gamma}(\beta) + (1 - k) \bar{v}(\gamma) \right\}$$

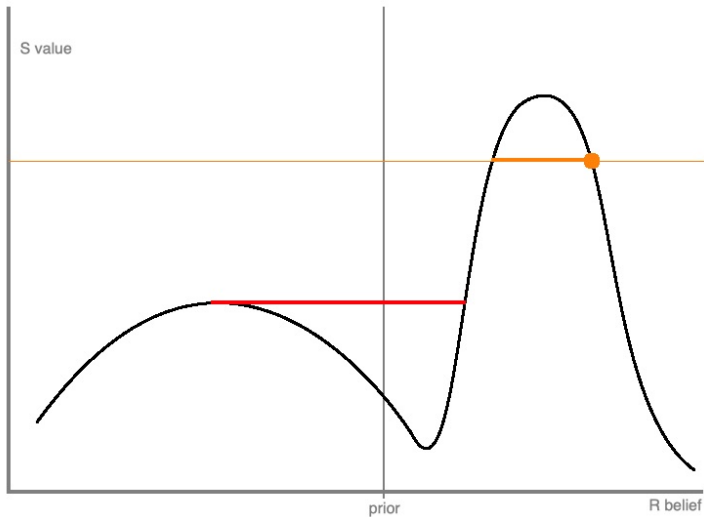
s.t. $k\beta + (1 - k)\gamma = \mu_0$

$$(1 - k)\gamma(\cdot) \geq (1 - \chi)\mu_0(\cdot)$$

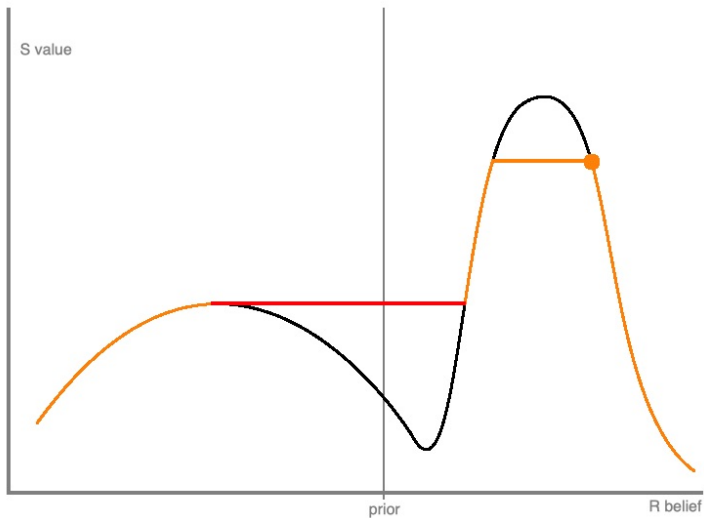
CAP THEN CONCAVIFY



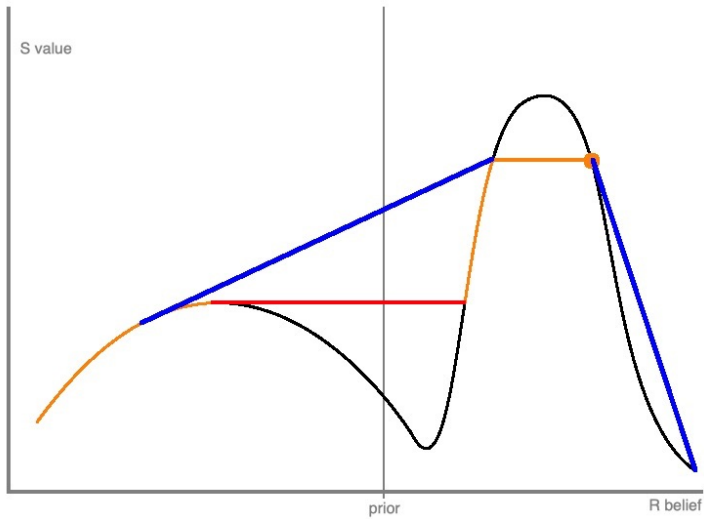
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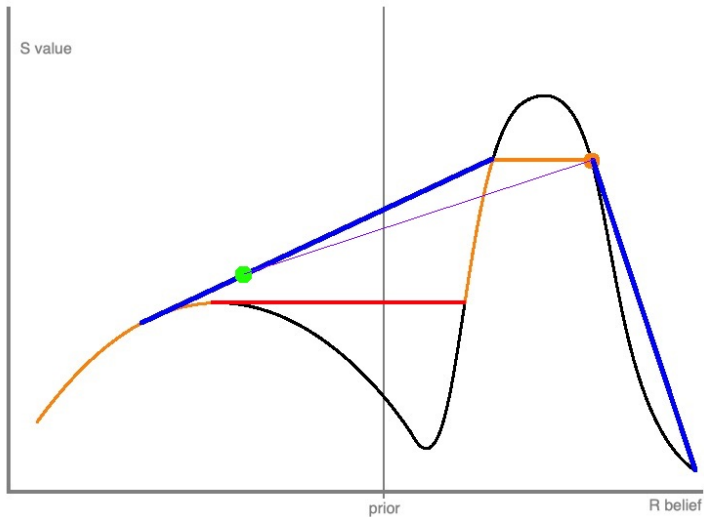
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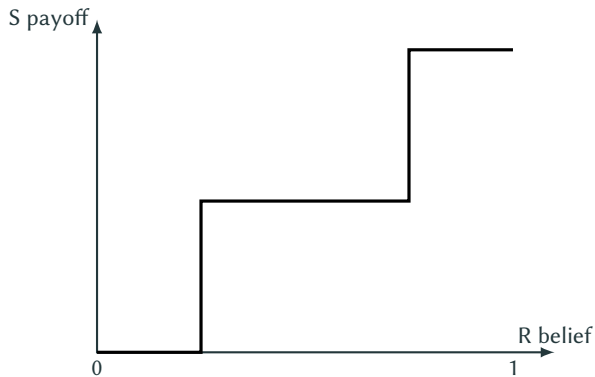
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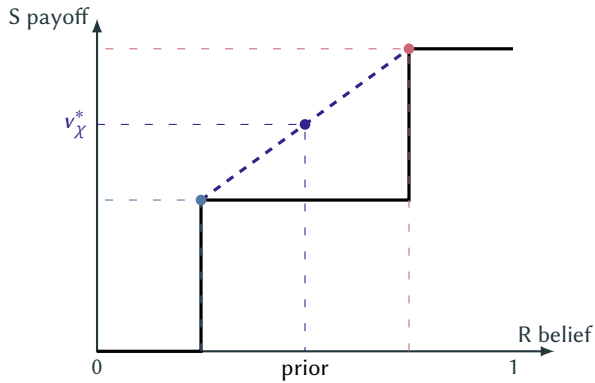
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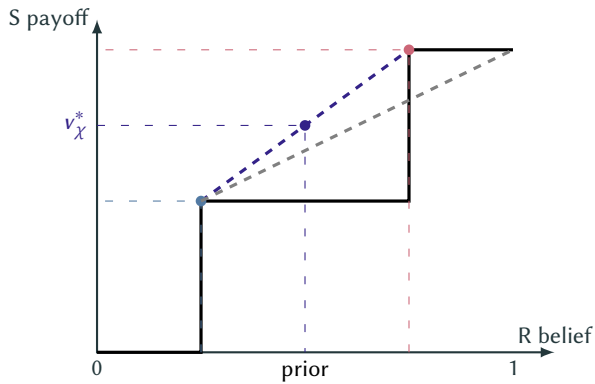
AN EXAMPLE: TWO STATES, THREE ACTIONS



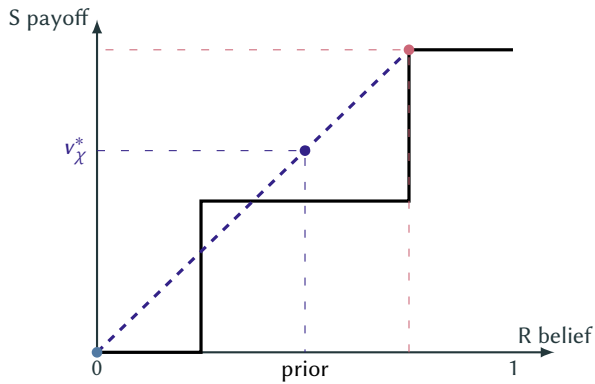
AN EXAMPLE: $\chi = 1$



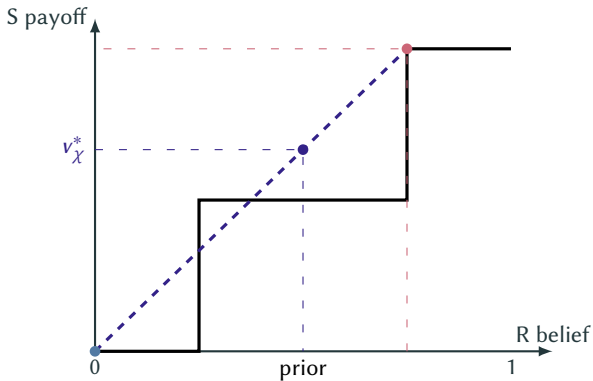
AN EXAMPLE: $\chi = \frac{3}{4}$



AN EXAMPLE: $\chi = \frac{2}{3}$



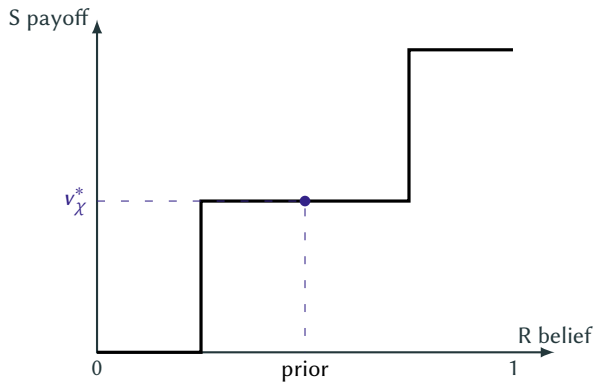
AN EXAMPLE: $\chi = \frac{2}{3}$



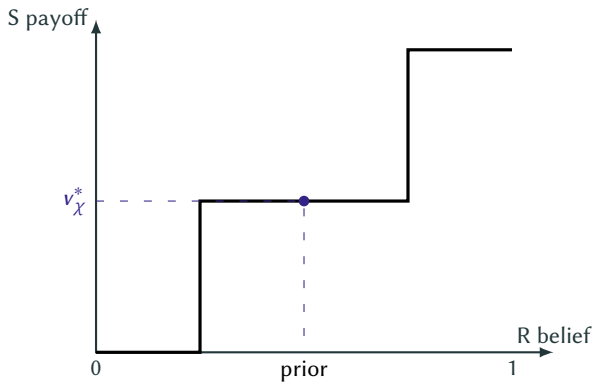
Productive mistrust

Lower credibility $\rightsquigarrow$ Better information

AN EXAMPLE: $\chi = \frac{2}{3} - \varepsilon$



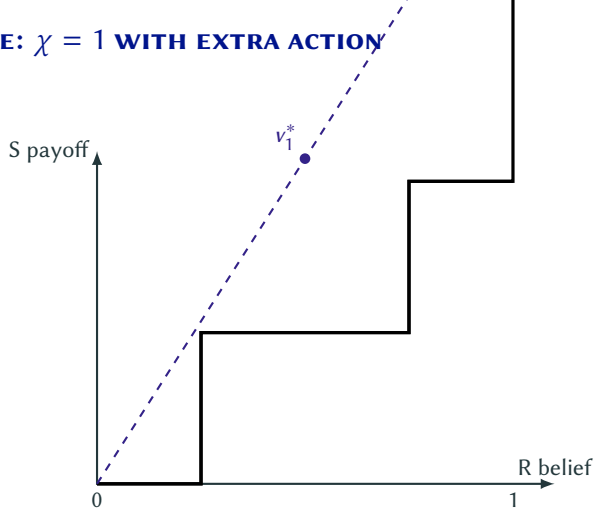
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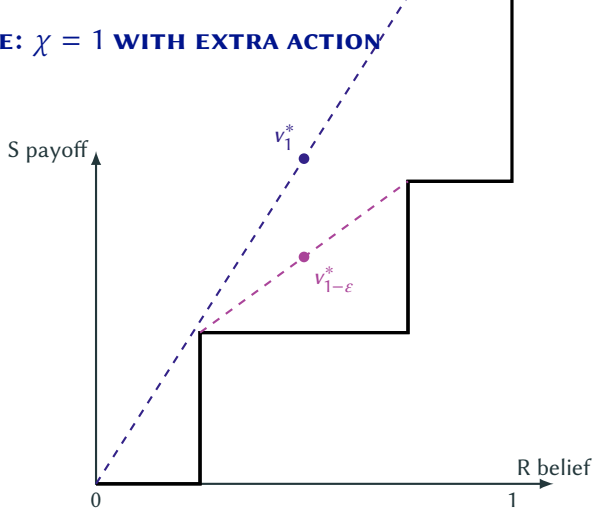
Collapse of trust

Slightly lower credibility $\rightsquigarrow$ Sender much worse off

AN EXAMPLE: $\chi = 1$ WITH EXTRA ACTION



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Fragility of commitment solution

Almost perfect credibility $\approx$ perfect credibility

SUMMING UP: STOCHASTIC COMMITMENT

Credibility shapes communication

Interpolated model: **degree of credibility** has delicate effects

Tools from information design still useful

Manipulable correlation

CORRELATION-MANIPULATION ROBUSTNESS

R sees message, but can't observe generating algorithm

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Credibility notion: robustness to **correlation manipulation**

- Valid reduced form for “many issue” interpretation

Say disclosure policy (with R response) is **credible** if

- S doesn't want to **change joint distribution** of states/messages
- ...while holding **marginal message distribution** fixed

CREDIBLE PERSUASION: LIN & LIU 2024 (EC '22)

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- $IC_S \iff$ cyclical monotonicity
- Every supported $\{(a_i, \theta_i)\}_{i=1}^n$ have (for $a_0 := a_n$)

$$\sum_{i=1}^n \left[u_S(a_i, \theta_i) - u_S(a_{i-1}, \theta_i) \right] \geq 0$$

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- Cyclical monotonicity $\iff$ monotonicity
- u_R supermodular and binary actions $\implies$ Bayesian persuasion
- u_R submodular and no u_R -duplicate actions $\implies$ no info

Mediated communication

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Attainable outcomes a polytope $\subseteq \Delta(A \times \Theta)$

- S prefers to report θ when type is θ
- R prefers to play a when recommendation is a
- Marginal on Θ is μ_0

Just correlated equilibrium if $|\Theta| = 1$

MEDIATION: A BELIEF-BASED APPROACH

Work with $p \in \Delta\Delta\Theta$ and belief-contingent R choice.

Key observation: Can infer $p_\theta(\mu) = \frac{\mu(\theta)}{\mu_0(\theta)}p(\mu)$.

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Salamanca 2021

- Replace IC_S constraints with associated Lagrangian term
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Corrao Dai 2023 (EC '23): state-independent S preferences

- Bayes-plausible $P \in \Delta(\text{gr} V)$ with **zero covariance** between μ, s
- Linear when fixing the prior: $\mathbb{E}[(\mu - \mu_0)s] = 0$
- **Mediation** $\sim$ **BP** $\iff$ **cheap talk** $\sim$ **BP**

Long cheap talk

DECENTRALIZED SENDER-RECEIVER PROTOCOLS

Mediated communication **requires some commitment** power

- Trusted third party
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- **Unrestricted, repeated, bilateral communication before decision**
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Bilateral communication helps

- R messages serve as **randomization**

Historical note

- Key tools from undiscounted repeated games (e.g., Hart 1985)

LONG CHEAP TALK: A BELIEF-BASED APPROACH

Aumann Hart 2003: name correspondence $\mathcal{E}^+ : \Delta\Theta \rightrightarrows \mathbb{R}^\Theta$ s.t. TFAE

- $s_0 \in \mathbb{R}^\Theta$ is an equilibrium S-payoff vector
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- Can also study static communication using $\mathcal{E}^+$
- s attainable $\iff$ some Bayes-plausible p has $p\{\mathcal{E}^+ \ni s\} = 1$

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Lipnowski Ravid 2020

- With transparent motives, long cheap talk doesn't help S

Other approaches

SOME MORE MODELS OF IMPERFECT-COMMITMENT COMMUNICATION

Informed information design

- Perez-Richet 2014, Koessler Skreta 2023

Lying costs

- Kartik 2009, Guo-Shmaya 2021, Perez-Richet Skreta 2021, Eilat-Neeman 2023

Repeated cheap talk

- Margaria Smolin 2018, Kuvalekar Lipnowski Ramos 2022, Best Quigley 2024, Mathevet Pearce Stacchetti 2024

Verifiable disclosure

- Huge, active literature

CONCLUDING REMARKS

Communication by informed parties is a key source of information

How much commitment power a sender has shapes what's possible

Variety of different models that capture “partial commitment”

Belief-based approach a powerful language for many of them

Thanks!

